

ARITHMETIC SATELLITES OF THE BRAGG COMB OF THE RIEMANN ZERO LATTICE: RESIDUE CLASSES, CHARACTERS, AND A BLIND TEST OF THE BOGOMOLNY–KEATING MIRROR LAW

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ABSTRACT. Companion notes found that the per-gap response of the Riemann zeros to the prime waves of the explicit formula is partly cancelled by the waves beyond any finite truncation, through a kernel concentrated on the Bragg comb of the lattice of gap midpoints. We study the satellites of that comb. Their positions form a height-independent grid $\Delta\omega = \omega - \log(t/2\pi) = \log(a/b)$. A stationary-phase analysis predicts that a satellite reads the residue class of the contributing prime power modulo a : the naive prediction of rotating class phases fails, while the refined law — class shares $\cos(2\pi rb/a)/\mu(a)$ — passes a blind test at $+\log 10$ (two other satellites follow its signs but exceed its magnitudes). On the zeros of real Dirichlet L -functions, satellites involving a prime that divides the conductor vanish, and the cancellation keeps the sign of the zeta case for both parities. The Euler product of the Bogomolny–Keating pair correlation gives the satellite amplitudes in closed form. It implies a mirror law — on the negative side the envelope is exactly $1/x$ at $\Delta\omega = -\log x$ — which a pre-registered blind test in a previously unexamined band confirms at 1.023 ± 0.023 of the prediction, excluding a line-density rival by more than 12σ . Coefficients with a positive exponent are transferred with a factor below one that relaxes towards one with height (exponent 1.36 ± 0.21 over three windows; a constant is excluded at 6.4σ); since the ratios-conjecture pair correlation has no such term, this is a property of our mixed zero–prime observable. Nothing here bears on a proof of the Riemann Hypothesis.

1. INTRODUCTION

The explicit formula writes the fluctuating part of the zero counting function as a sum of prime waves,

$$(1) \quad N(t) = \bar{N}(t) + S(t), \quad S(t) = - \sum_q a_q \sin(\omega_q t), \quad a_q = \frac{\Lambda(q)}{\pi \sqrt{q} \log q}, \quad \omega_q = \log q,$$

the sum running over prime powers q . Companion notes [1, 2, 3, 4] measured how the local statistics of the zeros respond to these waves: an amplitude channel and a zero-displacement channel, each a one-variable law in $\tau = \log q/L$, $L = \log(t/2\pi)$ [2, 3], and — in [4] — the diffraction pattern of the lattice of gap midpoints, with arithmetic spots at $\log p^k$ and Bragg combs at multiples of the mean density frequency $\omega = L$.

This note is about the neighbourhood of the first comb. Any measurement of the response to the waves must truncate the sum (1) at some τ_c ; the waves beyond the truncation do not simply add noise but *cancel* a fixed fraction of the interference (“mixing”) among the waves inside it, at a phase of exactly 180° . Resolving that cancellation by the frequency ω' of the cancelling waves gives a kernel $\kappa(\omega')$, and the kernel turns out to be concentrated on the comb $\omega' = L$ and on a family of *satellites* $\omega' = L + \log(a/b)$. The satellites carry arithmetic:

which of them light up, with which amplitude and with which internal structure is decided by the Möbius function, by residue classes, by Dirichlet characters and, as we show, by the Euler product of the Bogomolny–Keating pair correlation [5, 6, 7, 8].

Satellites at $\log(a/b)$ are the fingerprint, in the zero lattice, of the arithmetic terms of the pair correlation that are conjectural in the regime $\alpha > 1$ of Montgomery’s form factor [9, 10]; the single-point analogue is the Landau–Gonek formula for $\sum_{\gamma} x^{i\gamma}$ [11, 12, 13, 14], and the structure factor of the primes themselves has Bragg peaks with heights $\mu(q)^2/\varphi(q)^2$ at rational frequencies [15, 16]. We do not claim that the mechanism is new; what the note adds is a measurement programme on the zero side, organised as a sequence of pre-registered tests in which each prediction, its thresholds and its death conditions were frozen and published before the data were looked at, and in which failed predictions are reported next to the successful ones. We call a test *pre-registered* when the predictions, thresholds and analysis definitions were fixed and published (with a time stamp) before the relevant quantity was computed, and *blind* when, in addition, the data region it uses had never been displayed or searched. Results found after looking at the data are marked *post hoc*.

Section 2 defines the observable. Section 3 shows that the satellite positions are height-independent. Section 4 derives the selection rules from stationary phase and tests the residue-class law. Section 5 carries the analysis to Dirichlet L -functions. Section 6 derives the amplitudes from the Bogomolny–Keating Euler product and reports the blind test of the mirror law. Section 7 measures the height dependence of the positive-exponent coefficients, and Section 8 lists open problems.

2. THE OBSERVABLE

Data. We use A. M. Odlyzko’s table of the first 2×10^6 zeros [18, 17] and three windows of consecutive zeros: *alt*, the zeros 2×10^4 – 2×10^5 (1.8×10^5 gaps, mean $L = 9.34$; used only in Section 7); *low*, the zeros 2×10^5 – 5×10^5 (3×10^5 gaps, $L = 10.48$, $t \in [1.40, 3.19] \times 10^5$); and *last*, the top 3×10^5 gaps of the table ($L = 12.03$).

The per-gap identity. Let t_n be the ordinates, $g_n = t_{n+1} - t_n$ the gaps and $m_n = (t_n + t_{n+1})/2$ the midpoints. Differencing (1) over one gap, $N(t_{n+1}) - N(t_n) = 1$, gives for the unfolded gap deviation

$$(2) \quad \delta_n := \frac{g_n L(m_n)}{2\pi} - 1 = \sum_q 2a_q \sin\left(\frac{\omega_q g_n}{2}\right) \cos(\omega_q m_n),$$

up to the curvature of $\tilde{N}$ over one gap. Every wave enters the gap through its difference across the gap, sampled at the midpoint.

Truncation, mixing and cancellation. Truncate (2) to the *window* $\tau_q \leq \tau_c = 0.86$ and project the truncated series onto each window line, $c_q = 2\langle \delta_n^{(\tau_c)} e^{-i\omega_q m_n} \rangle_n$. The projection splits into the line’s own contribution and a *mixing* term, $c_q = c_q^{\text{self}} + \text{mix}_q$, produced by the other window lines. On a uniform grid distinct frequencies would be orthogonal; here the waves are sampled at the midpoints m_n and through the gaps g_n , which are themselves displaced by the primes, so the Gram couplings between wave columns are values of the midpoint structure factor $\hat{G}(\omega_q - \omega_{q'})$ [4] and do not vanish. The lines beyond the window

add a further projection Δ_q . Regressing it on the mixing over a pool of lines,

$$(3) \quad \zeta = \frac{\sum_{q \in \mathcal{P}} \Delta_q \overline{\text{mix}_q}}{\sum_{q \in \mathcal{P}} |\text{mix}_q|^2}, \quad \mathcal{P} = \{q : \tau_q \in [0.45, 0.86)\},$$

gives $\zeta = 0.333 \angle 180^\circ$ in the low window and $0.329 \angle 180^\circ$ in the last: the waves beyond the window cancel a third of the in-window mixing, at exactly opposite phase, independently of height.

The kernel. Resolve Δ_q by the frequency ω' of the lines that produce it. The window is cut into blocks b of consecutive gaps with mean L_b ; each line q' is assigned, in each block, to the slice $j = \lfloor (\omega_{q'} - L_b)/0.025 + \frac{1}{2} \rfloor$ of the *block-local* axis $\Delta\omega = \omega' - L_b$; the slice's per-gap series $\delta_n^{(b,j)} = \sum_{q' \in j} 2a_{q'} \sin(\omega_{q'} g_n/2) \cos(\omega_{q'} m_n)$ is projected onto the pool lines within the block, $C_{b,j}(q) = 2 \langle \delta_n^{(b,j)} e^{-i\omega_q m_n} \rangle_{n \in b}$, and

$$(4) \quad K_{b,j} = \frac{\sum_{q \in \mathcal{P}} C_{b,j}(q) \overline{\text{mix}_q}}{\sum_{q \in \mathcal{P}} |\text{mix}_q|^2}, \quad \kappa_j = - \frac{\sum_b n_b \text{Re } K_{b,j}}{\sum_b n_b},$$

the sums over b running over the blocks that cover slice j entirely so that $\kappa > 0$ is cancellation-directed and $\sum_j K_j$ reproduces the beyond-window part of ζ (to machine precision). For slices inside the window the slice's own contribution is removed from the truncated series before the mixing is formed, which makes the definition identical to the beyond-window one. Uncertainties are jackknife errors over eight contiguous groups of blocks; z denotes a signed ratio to that error and “ $n\sigma$ ” its magnitude. Where a decision rule uses σ_{eff} , the jackknife error has been inflated by a factor (1.2–1.6) calibrated, before the data were seen, on synthetic replicas processed by the same pipeline.

Line shape. A satellite at $\Delta\omega = \log r$ is produced, gap by gap, at $\omega' = L(m_n) + \log r$; within a block it is therefore smeared over the spread of $L(m_n)$. With eight blocks per window this spread is 0.07–0.15, which is larger than the spacing of many satellites; with 32 blocks, or blocks of equal width $\Delta L \leq 0.03$ (used from Section 6 on), it is 0.017–0.041. The line shape $h(j; r)$ follows parameter-free from the distribution of $L(m_n)$ in each block, and all amplitude measurements of Sections 6 and 7 are linear fits of the profile κ_j with these shapes and a linear baseline.

3. POSITIONS: A HEIGHT-INDEPENDENT GRID

Mapped against the frequency of the cancelling lines and the pool band, the kernel has a simple structure. It is 77% rank-one (a pre-registered threshold of 80% for a rank-one mechanism was not met), with a band factor at $180^\circ \pm 5^\circ$ in every pool band, and its structure sits on a comb in τ' : the Bragg line $\tau' = 1$ and satellites; the strongest feature beyond the Bragg line, at $\tau' \in (1.05, 1.10]$, is 96% carried by the satellites of 2 and 3.

The natural axis is not τ' but $\Delta\omega$.

Observation 1 (Height independence; pre-registered). *In the low window ($L = 10.48$), with the definitions frozen on the last window ($L = 12.03$), the block-median kernel peaks fall at $\Delta\omega = 0.025, 0.725, 1.113$ and 1.800 , against the targets $0, \log 2 = 0.693, \log 3 = 1.099$ and $\log 6 = 1.792$, all within ± 0.05 , and far from the positions $0.604, 0.957, 1.562$ predicted by a fixed- τ' comb; the block-to-block slopes are zero, where a fixed- τ' reading requires -1 .*

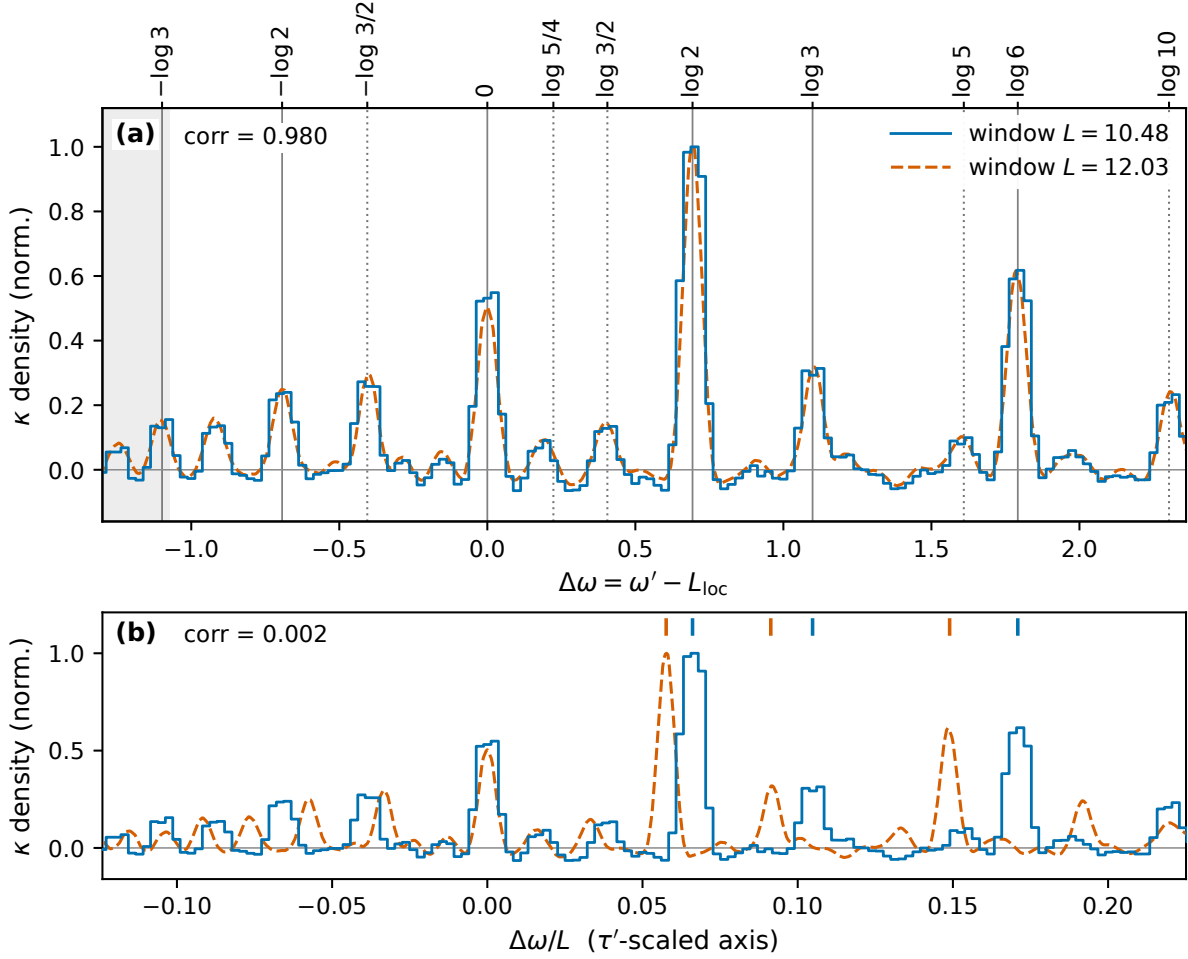


FIGURE 1. **Satellite positions are universal in $\Delta\omega$.** (a) Pooled cancellation-kernel density κ , normalized to its maximum on $-1.3 \leq \Delta\omega \leq 2.3$, versus $\Delta\omega = \omega' - L_{\text{loc}}$ for two windows of zeros, $L = 10.48$ (solid, bins of 0.025) and $L = 12.03$ (dashed, τ' bins of width 0.060 in ω); vertical lines mark satellites at $\log(a/b)$, solid for the pre-registered targets $0, \pm \log 2, \pm \log 3, \log 6$ and dotted for the other satellites that are bright in both windows. In the $L = 10.48$ window the pre-registered block-median peaks lie at 0.025, 0.725, 1.113 and 1.800 (targets $0, \log 2, \log 3, \log 6$), each within 0.05 of its target and away from the τ' -fixed alternatives 0.604, 0.957, 1.562, and the two profiles correlate at 0.980. (b) On the τ' -scaled axis $\Delta\omega/L$ the same profiles no longer coincide (correlation 0.002; ticks mark $\log n/L$ for $n = 2, 3, 6$ in each window). The shaded strip $\Delta\omega < -1.07$ is only partly covered by the eight blocks of the $L = 10.48$ window.

Post hoc, the two windows' profiles overlap on the $\Delta\omega$ axis with correlation 0.980, against 0.002 on the τ' -scaled axis.

Figure 1 shows the overlap. The cancelling waves are those whose frequency exceeds the local density frequency by the logarithm of a rational number.

4. SELECTION RULES AND RESIDUE CLASSES

Derivation 2 (Stationary phase). *The midpoints satisfy $N(m_n) = n$ exactly, with $2\pi\bar{N}(t) = tL(t) - t + 7\pi/4$. Hence $e^{i\omega'm_n} = e^{i(\omega'm_n - 2\pi\bar{N}(m_n))} e^{-2\pi iS(m_n)}$, and by the Jacobi–Anger expansion $e^{-2\pi iS} = \prod_q \sum_k J_k(2\pi a_q) e^{ik\omega_q t}$. The term labelled $\{k_q\}$ has phase $\phi(t) = (\omega' + \sum_q k_q \omega_q)t - 2\pi\bar{N}(t)$ with a stationary point at $L(t^*) = \omega' - \log n$, $n := \prod_q q^{-k_q}$, which may be rational. Satellites therefore sit at $\Delta\omega = \log n$, for every height. Writing $n = a/b$ in lowest terms, $t^* = 2\pi q'b/a$ for a line q' with $\omega' = \log q'$, and the stationary value of the phase is $2\pi q'b/a$ up to the constant $-7\pi/4 - \pi/4 \equiv 0$ (the last $-\pi/4$ is the stationary-phase factor of a phase with $\phi'' = -1/t < 0$). The contribution of a line depends on $q' \bmod a$.*

Two predictions follow. (a) *Selection*: prime powers are equidistributed over the reduced residue classes modulo a , so the coherent sum over lines is a Ramanujan sum, $c_a(1)/\varphi(a) = \mu(a)/\varphi(a)$ [19]; satellites with $\mu(a) = 0$ vanish. (b) *Class structure*: empirically the kernel follows the real part of the midpoint structure factor at ω' (a proxy that explains 90–95% of it), so it reads the real part of the class phase, and the share of residue class r in the satellite is

$$(5) \quad s_r = \frac{\cos(2\pi r b/a)}{\mu(a)}.$$

The first version of this analysis (without the final $-\pi/4$, and with the class *phases* rather than their real parts as observable) predicted that the classes of $+\log 3$ would appear 120° apart. That prediction was pre-registered and *failed*: the two classes of $+\log 3$ are $-8.6^\circ \pm 3.9^\circ$ apart and carry 0.46 and 0.54 of the satellite, statistically indistinguishable from the control $-\log 3$ ($-0.9^\circ \pm 5.5^\circ$). The selection rule survived: none of nine positions with $\mu(a) = 0$ lights up (the largest is $+1.6\sigma$), while all six main satellites do (7.6 – 12.4σ); only three of the nine are, however, consistent with zero — the others carry significant *negative* values. The refined law (5) was then pre-registered as a blind test on a satellite never analysed by class.

Observation 3 (Residue classes of $+\log 10$; blind). *The four classes $r = 1, 9, 3, 7 \pmod{10}$ of the satellite $+\log 10$ (last window) carry the shares $+0.899 \pm 0.023$, $+0.902 \pm 0.017$, -0.416 ± 0.035 , -0.386 ± 0.035 , against $\cos(2\pi r/10) = +0.809, +0.809, -0.309, -0.309$: all four signs, at $|z| = 11$ – 52 , each within 0.11 of the prediction. The controls behave as required: $+\log 3$ splits 0.462/0.538 and $-\log 3$ (with $a = 1$, no class structure) 0.499/0.501.*

The class structure of $+\log 7$ has the predicted signs in all four classes ($|z| = 4.9$ – 6.2) but magnitudes 1.8–2.4 times the prediction; $+\log 5$, reported without verdict, shows a smaller stretch (1.25–1.75). We do not know its origin. A pre-registered attempt to explain it by a negative kernel floor between satellites failed: away from satellites the kernel is consistent with zero ($+0.00024 \pm 0.00017$). The positions with $\mu(a) = 0$ are not exactly zero either: they carry small negative residues, 9.4σ below the empty-window floor in the last window, and of the same size in all three windows of Section 7.

5. DIRICHLET L -FUNCTIONS

For a primitive real character χ modulo k the counting function has $L_\chi(t) = \log(kt/2\pi)$ and waves weighted by $\chi(q)$. The Jacobi–Anger factor of a wave with $\chi(q) = 0$ is trivial, so satellites $\log(a/b)$ with a prime of ab dividing k are absent. The stationary point moves to $t^* = 2\pi q'b/(ka)$, the class sum becomes $\chi(a)\bar{\chi}(b)\tau(\chi)\mu(a)$ with the Gauss sum $\tau(\chi)$ (not to be confused with $\tau = \log q/L$), the character signs cancel against the amplitudes $\chi(a)\chi(b)$, and

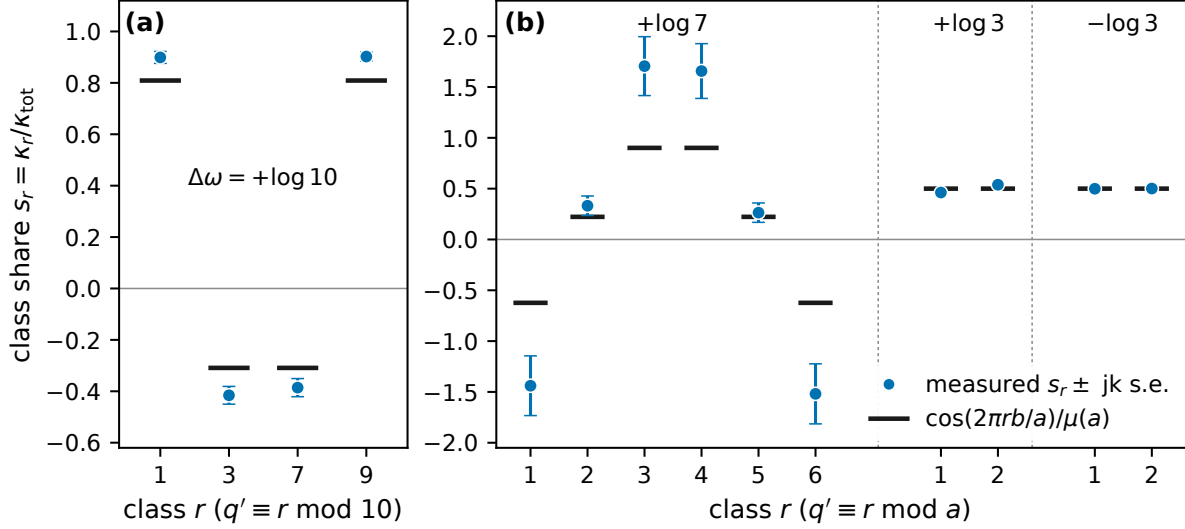


FIGURE 2. **Residue-class law of the satellites.** Class shares $s_r = \kappa_r / \kappa_{\text{tot}}$ of the lines $q' \equiv r \pmod{a}$ (points, 8-block jackknife errors) against the blind prediction $s_r = \cos(2\pi r b / a) / \mu(a)$ (bars), window $L = 12.03$. (a) At $\Delta\omega = +\log 10$ all four classes have the predicted sign and lie within 0.11 of the prediction: $s_1 = 0.899 \pm 0.023$ and $s_9 = 0.902 \pm 0.017$ (predicted 0.809), $s_3 = -0.416 \pm 0.035$ and $s_7 = -0.386 \pm 0.035$ (predicted -0.309). (b) At $+\log 7$ the signs are again correct ($|z| = 4.9$ – 6.2 for $r = 1, 3, 4, 6$) but the magnitudes are 1.8–2.4 times the prediction, while the controls $+\log 3$ ($0.462/0.538$) and $-\log 3$ ($0.499/0.501$, no class structure expected) sit at $\frac{1}{2}/\frac{1}{2}$.

the phase of $\tau(\chi)$ (real for even, imaginary for odd χ) combines with the parity-dependent constant of $\bar{N}_\chi$ to give a *real* kernel with the sign of the zeta case for both parities. Two rival pictures — one that forgets the conductor in t^* (no cancellation on the islands) and one that forgets the parity (no cancellation for odd χ) — were pre-registered alongside.

Observation 4 (Characters; pre-registered). *On the zeros of $L(s, \chi)$ for χ_4, χ_3, χ_5 (even), χ_8 (even) and χ_8 (odd) up to $t \approx 5 \times 10^4$: all 21 satellites forbidden by the conductor are dark in the structure factor (22 of the 24 allowed ones at $+9.8$ to $+12.1\sigma$, the two exceptions being $+\log 4$, where $\mu(4) = 0$) and suppressed in the kernel to a small residue; the allowed satellites cancel on all five islands, for both parities, with the zeta sign and a real phase ($180^\circ \pm 1.3^\circ$; $z = -55$ to -82 for the primary satellite pairs); the even and odd characters modulo 8 agree to 1.009 ± 0.024 . Both rivals die.*

Two parts of the prediction failed. The forbidden satellites leave a residue of 3–5% of the allowed ones, with opposite sign — the same level as the $\mu(a) = 0$ residues of the zeta case. And the predicted amplitude ratio to zeta at matched L , $\sqrt{k}/\varphi(k)$, is wrong by factors 1.5–4.5; a variant with $(k/\varphi(k)) \prod J_0$ recorded before the data but outside the verdict is within 3–16% in six pairs.

6. THE BOGOMOLNY–KEATING MIRROR LAW

The first-order amplitude rule of Derivation 2 — brightness $\propto |\mu(n)|/\varphi(n)$ on the positive side and $\propto 1/m$ at $n = 1/m$ — gets the ordering of the satellites right but not their

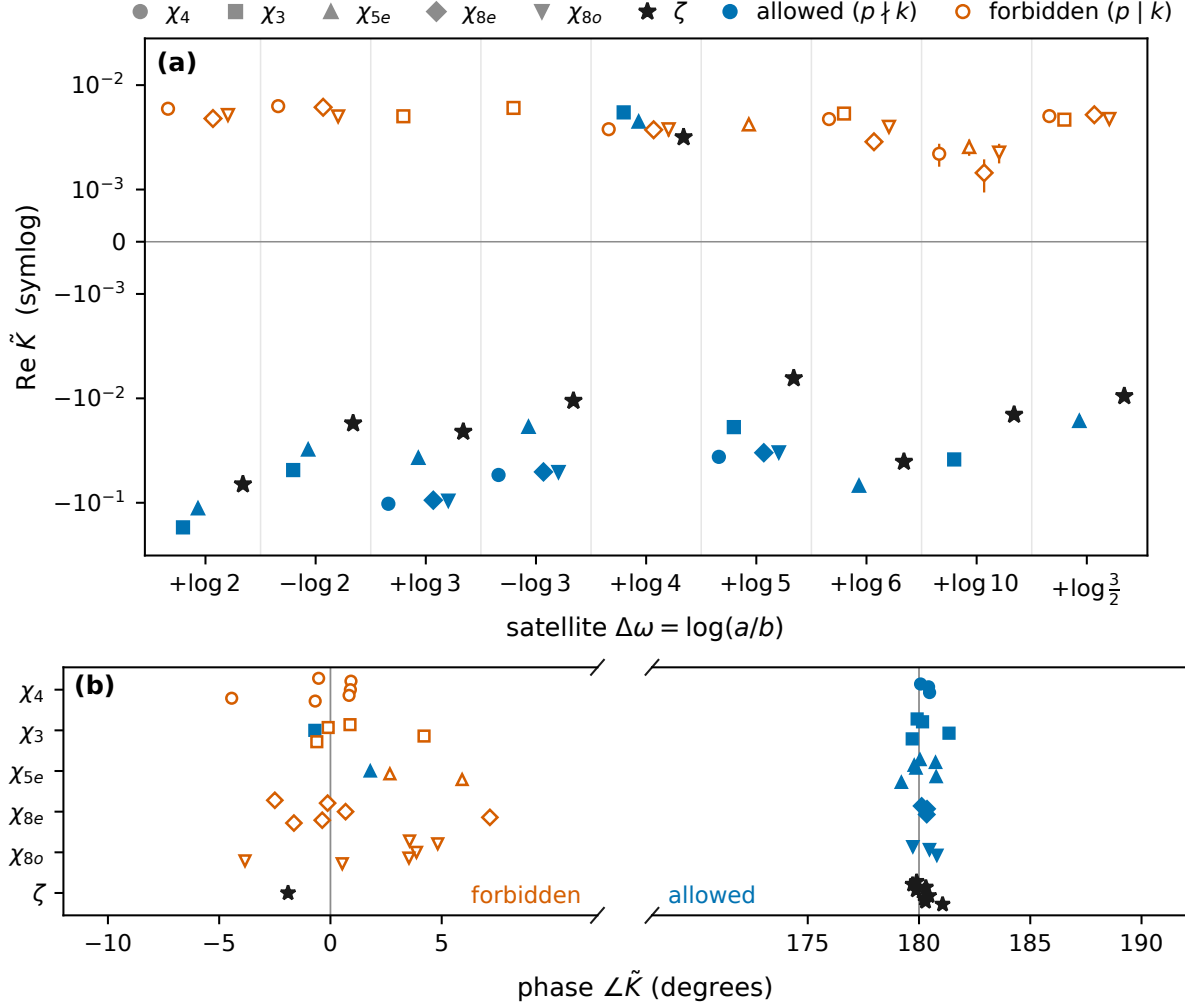


FIGURE 3. **The character selects the satellites of Dirichlet L -function zeros.** (a) $\text{Re } \tilde{K}$ (symlog scale, jackknife errors) at nine satellites for five islands of zeros ($\chi_4, \chi_3, \chi_{5e}, \chi_{8e}, \chi_{8o}$; $L_\chi \geq 8.5$) and for ζ (stars; low window on the same ΔL block grid); filled symbols are allowed satellites ($n = a/b$ has no prime factor $p \mid k$), open symbols forbidden ones. Allowed satellites with $\mu(a) \neq 0$ carry the sign of ζ in both parities ($z = -55$ to -82 for the six primary pairs; $\chi_{8o}/\chi_{8e} = 1.009 \pm 0.024$ at $+\log 3$), whereas forbidden satellites leave only a small residual of opposite sign, $+0.0014$ to $+0.0063$, at the level of the $\mu(4) = 0$ position $+\log 4$ (positive also for ζ , χ_3 and χ_{5e}). (b) Phases: $\angle \tilde{K} = 180^\circ \pm 1.3^\circ$ for every allowed $\mu \neq 0$ satellite, odd or even, and $0^\circ \pm 7.2^\circ$ for every forbidden one.

magnitudes: odd primes fade faster than predicted. The amplitudes are instead given by the Bogomolny–Keating pair correlation [5, 6, 7], re-derived from the ratios conjecture by Conrey and Snaith [8], whose off-diagonal part is, in the form of [7, eqs. (7.5)–(7.6)],

$$(6) \quad R_2^{\text{off}}(\varepsilon) \propto e^{iL\varepsilon} |\zeta(1+i\varepsilon)|^2 \Phi(\varepsilon) + \text{c.c.}, \quad \Phi(\varepsilon) = \prod_p \left[1 - \frac{(1-p^{i\varepsilon})^2}{(p-1)^2} \right].$$

The carrier $e^{iL\varepsilon}$ places the structure on the comb; the envelope $f = |\zeta(1 + i\varepsilon)|^2 \Phi$ is an Euler product, $f = \prod_p g_p(\varepsilon \log p)$ with $g_p(\theta) = |1 - e^{-i\theta}/p|^{-2} [1 - (1 - e^{i\theta})^2/(p-1)^2]$, and its Bohr coefficient at frequency $\log r$ (the coefficient of $e^{i\varepsilon \log r}$ in its almost-periodic expansion) is the amplitude of the satellite $\Delta\omega = \log r$.

Derivation 5 (Closed form). *The first factor of g_p has Fourier coefficients $A_k = p^{-|k|}/(1 - p^{-2})$; the second is a trigonometric polynomial with coefficients $B_0 = 1 - (p-1)^{-2}$, $B_1 = 2(p-1)^{-2}$, $B_2 = -(p-1)^{-2}$. Hence $\hat{g}_p(k) = \sum_j B_j A_{k-j}$, and*

$$\frac{\hat{g}_p(k)}{\hat{g}_p(0)} = \begin{cases} p^k, & k \leq 0, \\ p/(p-1)^2, & k = 1, \\ 0, & k \geq 2, \end{cases}$$

the last because $\hat{g}_p(k) \propto p^{-k}(B_0 + pB_1 + p^2B_2) = 0$. The satellite at $\log r$, $r = \prod_p p^{k_p}$ in lowest terms, has relative amplitude $c(r) = \prod_p f_p(k_p)$ with f_p the right-hand side.

Three consequences. *Mirror law*: on the negative side, $c(1/n) = 1/n$ for every integer n and $c(2/m) = 2/m$ for odd m , so the envelope at $\Delta\omega = -\log x$ is exactly $1/x$ on the half-integers; between them, satellites $x = b/3$ and $x = b/6$ lie on a quarter envelope $0.25/x$, and all other families are below 0.03. *Prime-power asymmetry*: $+\log p^k$ with $k \geq 2$ is exactly dark (as the selection rule of Section 4 says), but $-\log p^k$ is bright. *Positive side*: $c(p) = p/(p-1)^2$, so $c(2) = 2$, $c(3) = 3/4$, $c(5) = 5/16$, $c(7) = 7/36$; the factor $f_2(1) = 2$ explains why $+\log 2m$ is twice as bright as $+\log m$ for odd m .

A lesson in line shape (post hoc). Compared with the 23 satellite amplitudes of the first catalogue (eight blocks, point readings, normalised to $+\log 2$), the closed form fits the mirror side ($-\log 2$: 0.96, $-\log 3$: 1.05 of the prediction) but misses several mixed ratios by 3–10 σ , one of them with the wrong sign ($+\log(5/3)$). With 32 blocks and profile fits these misses largely disappear ($+\log(5/3)$: $-0.44 \rightarrow +0.42$; $+\log(5/2)$: $0.17 \rightarrow 0.57$; $-\log(4/3)$: $0.31 \rightarrow 0.79$): they were produced by the eight-block line shape, whose width exceeds the satellite spacing.

The blind test. The mirror law was tested in the band $-2.12 \leq \Delta\omega \leq -1.58$ ($4.86 \leq x \leq 8.33$), whose part below -1.775 had never been computed and whose remainder had been computed in an earlier map but never displayed or searched (no figure or search had gone below -1.35). The test was pre-registered after an independent derivation audit; its design was changed several times before the data were seen, each time because an audit or a synthetic power study showed a weakness.

The final design fits the band with the seven main satellites free, the quarter-envelope satellites fixed from a calibration window on the seen side, and a linear baseline. It compares two hypotheses by self-consistency: *BK* (amplitudes s/x , with the scale s fixed by $-\log 2$ and $-\log 3$) and a rival in which the amplitude also carries the smooth line-density factor $e^{\omega'/2}/\omega'$ (predicting 0.71 of BK in the band). In 2×200 synthetic replicas the correct hypothesis won in 93% and 86% of cases and the wrong one in at most 0.5%.

Observation 6 (Mirror law; blind). *In the low window, the ratios $R_x = A_x/(s/x)$ of the integer family are 1.016 ± 0.018 , 1.014 ± 0.033 , 1.046 ± 0.031 , 1.048 ± 0.060 for $x = 5, 6, 7, 8$; their weighted mean is 1.023 ± 0.023 , and the rival is excluded at more than 12 σ . The prime power $x = 8$ lies on the envelope (0.993 ± 0.039 of the fitted $x^{-\beta}$ law), although $+\log 8$ is*

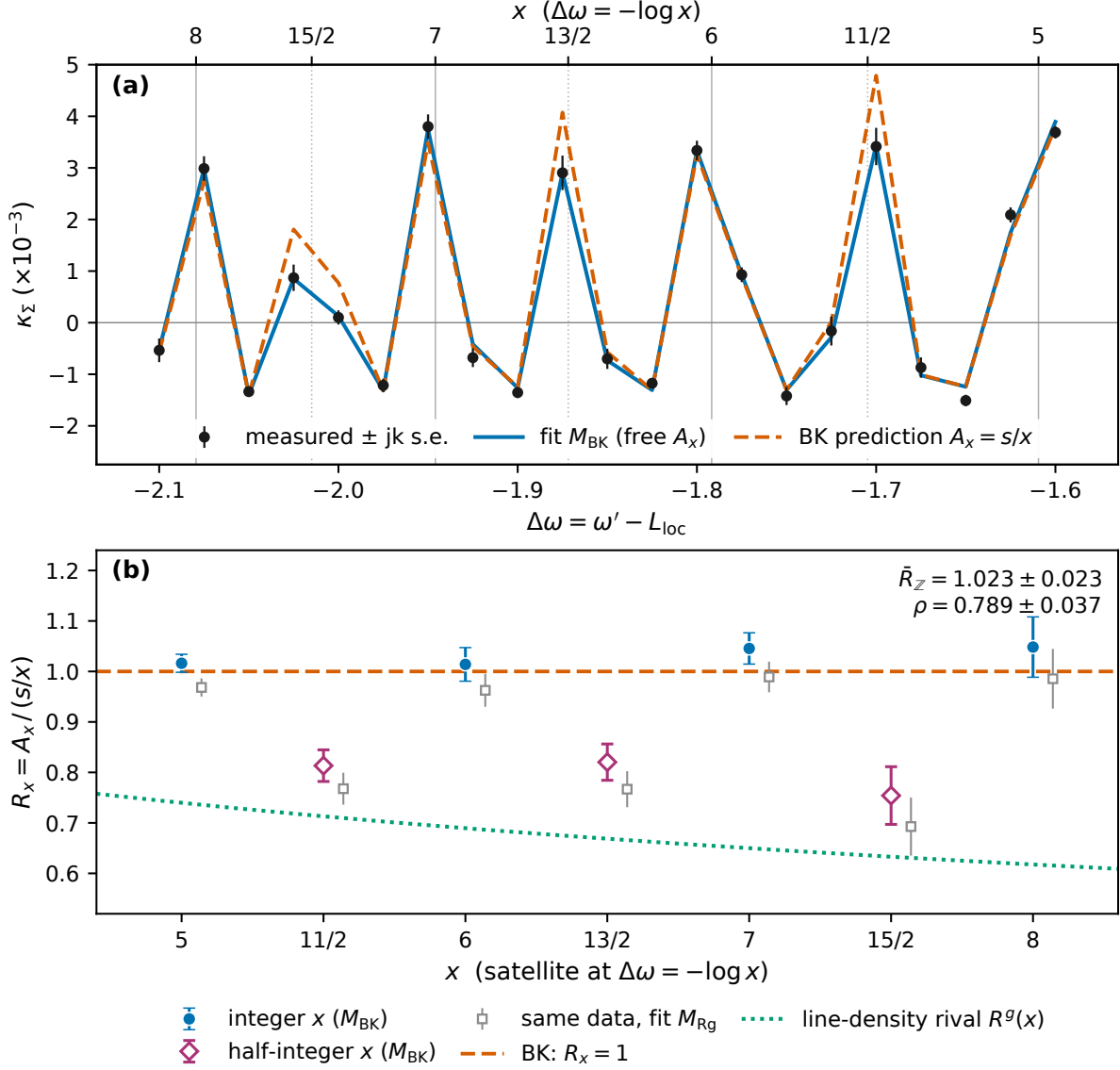


FIGURE 4. **Mirror law in the blind band.** (a) Kernel κ_Σ (32 blocks, window $L = 10.48$, jackknife errors) in the pre-registered, previously unviewed band $-2.12 \leq \Delta\omega \leq -1.58$, with the free-amplitude fit M_{BK} and the parameter-free Bogomolny–Keating prediction $A_x = s/x$ (s from the calibration band $-1.30 \leq \Delta\omega \leq -0.55$; same linear baseline and fixed quarter-family terms); solid and dotted grid lines mark integer and half-integer x at $\Delta\omega = -\log x$. (b) Amplitude ratios $R_x = A_x/(s/x)$: the integer family gives $\bar{R}_Z = 1.023 \pm 0.023$ (σ_{eff}), whereas the line-density rival predicts $\bar{R}^g = 0.706$ and its own fit (grey) returns 0.972, more than $12\sigma_{\text{eff}}$ away. The half-integer family follows the same $1/x$ envelope at a reduced level, $\rho = 0.789 \pm 0.037$, and the prime power $x = 8$ is not suppressed ($R_8 = 1.048 \pm 0.060$, $\psi = 0.993 \pm 0.039$).

dark. The half-integer family ($x = 5.5, 6.5, 7.5$) lights up at 0.789 ± 0.037 of the integer family — inside the pre-registered band $|\rho - 1| \leq 0.25$, but significantly below one.

7. POSITIVE EXPONENTS AND HEIGHT

Relative to the mirror scale of the same window, the positive-exponent coefficients are transferred with a factor below one. In the low window $\eta_p := A(+\log p)/(s c(p))$ is 0.79, 0.71, 0.60, 0.54 for $p = 2, 3, 5, 7$, and the same factors account, approximately multiplicatively, for the half-integer family (η_2) , the quarter-envelope families $(\eta_3, \eta_2\eta_3)$ and the mixed ratios. A post-hoc look suggested $\eta_p \approx p^{-1/3}$. Whether this is a constant or a finite-height effect was pre-registered as a test over three windows, with the model $1 - \eta_p(L) = B_p(L/L_0)^{-\gamma}$: $\gamma = 1$ for a correction of relative order $1/L$, $\gamma = 0$ for a constant.

window	L	η_2	η_3	η_5	η_7
alt (blind)	9.34	0.784(14)	0.663(13)	0.583(19)	0.521(28)
low	10.48	0.789(11)	0.713(15)	0.604(21)	0.538(20)
last	12.03	0.837(9)	0.770(7)	0.712(14)	0.616(26)

TABLE 1. Positive-exponent transfer factors in three windows (jackknife errors in the last digits). The alt window is fully blind; coarse satellite ratios of the last window had been seen before the test (see text).

Observation 7 (Height dependence; pre-registered). *The common exponent is $\gamma = 1.36 \pm 0.21$; a constant is excluded at 6.4σ . The ratio of the last to the alt window, which does not use the window in which the pattern was first seen, excludes a constant at 7.9σ (and is 1.9σ steeper than $\gamma = 1$). The half-integer families relax towards one in the same way ($R(5/2)$: $0.79 \rightarrow 0.83 \rightarrow 0.92$; $R(7/2)$: $0.75 \rightarrow 0.81 \rightarrow 0.88$), as does the quarter envelope ($0.65 \rightarrow 0.69 \rightarrow 0.75$), while the integer mirror side stays at one in all windows.*

Two caveats belong with this result. The decisive leverage comes from the last window, whose coarse ratios had been seen (without distinguishing the hypotheses) in an earlier test; the fully blind alt window alone gives 1.074 ± 0.055 against 1.122 ($\gamma = 1$) and 1 ($\gamma = 0$), as the power study had anticipated. And three windows cannot fix the functional form: the change is small between $L = 9.3$ and 10.5 and large between 10.5 and 12.0 . The $p^{-1/3}$ pattern was a coincidence of height.

The negative residues at the positions with $\mu(a) = 0$ (Section 4) do not relax: in units of the mirror scale they are -0.023 , -0.023 , -0.032 at $+\log 4$, -0.021 , -0.018 , -0.017 at $+\log 8$ and -0.020 , -0.016 , -0.020 at $+\log 9$ in the alt, low and last windows.

What does the height dependence mean? The Conrey–Snaith pair correlation contains all lower-order arithmetic terms, with an error $O(T^{1/2+\varepsilon})$ [8], and its carrier term has L -independent Bohr coefficients. The relaxation is therefore not a correction to (6); it is a property of the way our mixed zero–prime observable transfers the coefficients of R_2 — exactly on the negative side, and only asymptotically on the positive side.

8. OPEN PROBLEMS

- (1) *The transfer function.* Derive $\eta_p(L)$ from the definition (4) and the identity (2): why is the transfer exact for negative exponents and asymptotic for positive ones?

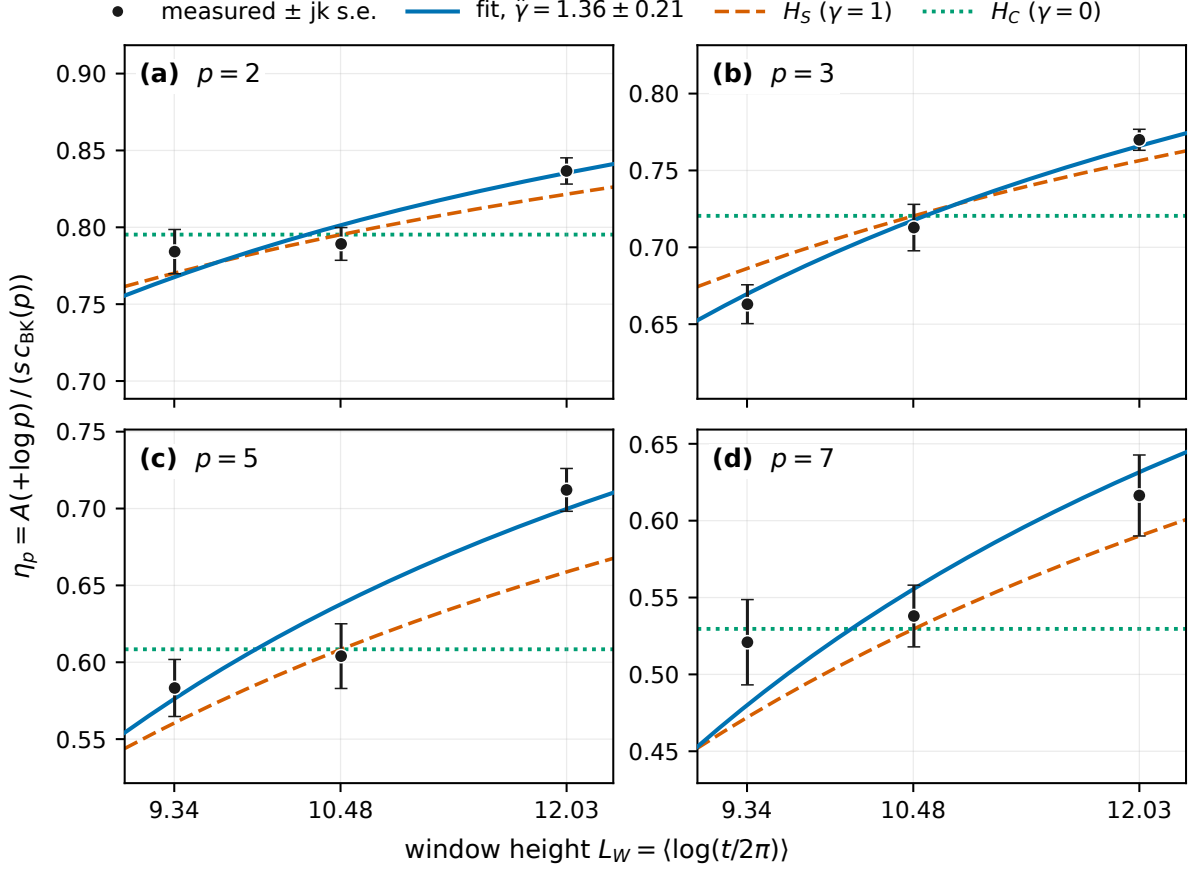


FIGURE 5. **Height dependence of the positive-exponent factors.** $\eta_p = A(+\log p)/(s c_{\text{BK}}(p))$ for $p = 2, 3, 5, 7$ in three windows, $L_W = 9.34$ (blind), 10.48 (reference) and 12.03, with jackknife errors. Solid: joint fit $1 - \eta_p = B_p(L/L_d)^{-\gamma}$ with $L_d = 10.48$, giving $\hat{\gamma} = 1.36 \pm 0.21$ (σ_{eff}); dashed: finite-height hypothesis H_S ($\gamma = 1$); dotted: constant structure H_C ($\gamma = 0$), both anchored at the pre-registered values $\eta_p = 0.795, 0.720, 0.608, 0.530$ (the first, 32-block measurement; the low-window values of Table 1 use equal- ΔL blocks). $\hat{\gamma}$ lies 1.7σ from H_S and 6.4σ from H_C : the factors relax towards the Bogomolny–Keating value $\eta_p = 1$ as the height grows.

- (2) *The $\mu = 0$ residues.* Positions where the closed form is exactly zero carry negative residues of about 2% of the mirror scale, independent of height, and the conductor-forbidden satellites of L -functions carry residues at the same level. Are they a term beyond (6)?
- (3) *Magnitudes.* The class shares of $+\log 7$ and $+\log 5$ exceed (5) by factors 1.25–2.4, and the L -function amplitudes follow $(k/\varphi(k)) \prod J_0$ rather than $\sqrt{k}/\varphi(k)$. Neither is derived.
- (4) *Half-integers.* The half-integer family sits at ≈ 0.8 of the mirror law and relaxes with height like η_2 ; is it η_2 ?

A recent preprint [20] reports no detectable arithmetic frequencies in the deviations of Δ_3 from random-matrix theory. We note that on the unfolded axis used there the arithmetic

frequencies sit at $2\pi \log p / \log(T/2\pi)$ rather than at $\log p$, which may account for the null result; the satellites reported here are, in any case, a different observable.

REPRODUCIBILITY

All code, data products, pre-registration files (with sha256 stamps and the dates at which they were published) and the research log, including the failed predictions, are available in the repository <https://github.com/azizran/riemann-zero-statistics> (directory `qm_riemann/`; Section 3: tasks 188, 190; Section 4: 192–194; Section 5: 195; Section 6: 196–197; Section 7: 198), archived at Zenodo, doi:10.5281/zenodo.22942475.

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